

Introduction to Network Devices

Task – 1 : Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

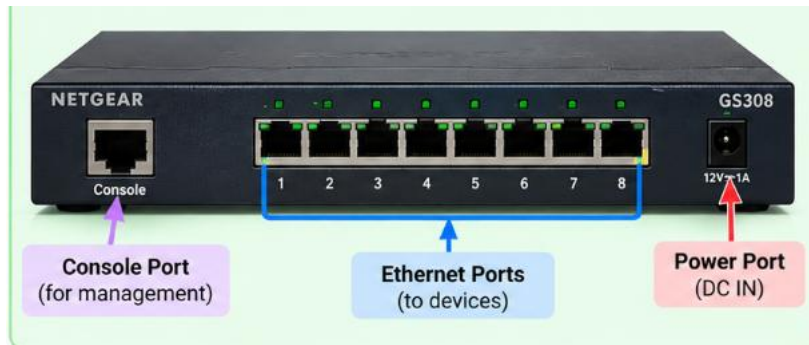
Solution : **Network Devices: Functions and Typical Ports**

Network Device	Main Function	Typical Ports / Interfaces	Use in a Home Wi-Fi Setup
Switch	Connects multiple wired devices within the same local network and forwards data to the correct device.	Ethernet ports (RJ-45), sometimes Console port, Power	Connects PCs, TVs, printers, or other wired devices to the router.
Router	Connects the local network to the Internet and routes packets between different networks.	WAN Ethernet, LAN Ethernet, sometimes USB, Power	Connects the home network to the ISP/Internet and usually provides DHCP and NAT.
Hub	Connects multiple devices and broadcasts incoming data to every connected port.	Ethernet ports (RJ-45), Power	Older networks; rarely used in modern home networks.
Firewall	Monitors and controls network traffic according to security rules.	Ethernet/WAN/LAN ports, Console/Management port, Power	Protects the home network from unwanted or unauthorized traffic.
Access Point (AP)	Provides wireless network access and connects Wi-Fi devices to the wired network.	Ethernet port, Power; antennas may be internal or external	Allows phones, laptops, tablets, and other Wi-Fi devices to connect to the network.

Task – 2 : Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

Solution : Identify Network Devices and Ports

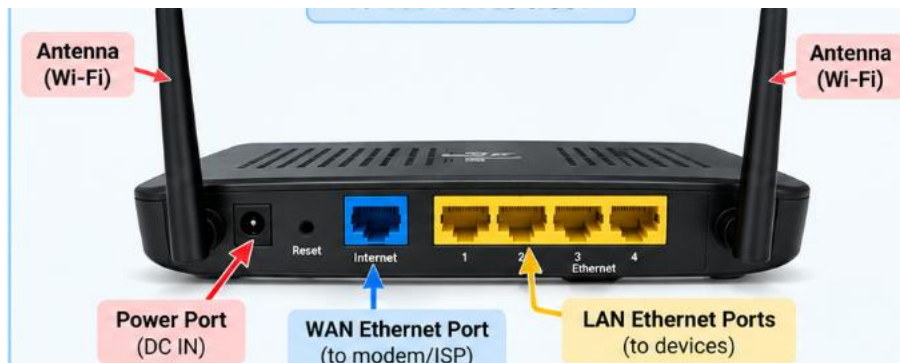
A. Switch



Important ports:

- **Ethernet/RJ-45:** Connects computers and other wired devices.
- **Console:** Used for device configuration and management.
- **Power:** Supplies electrical power.

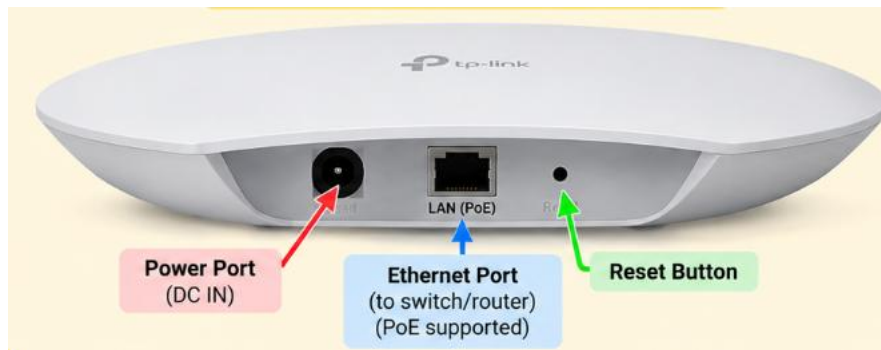
B. Router



Important ports:

- **WAN port:** Connects the router to the ISP/modem.
- **LAN Ethernet ports:** Connect wired devices.
- **Power:** Provides electrical power.
- **Antennas:** Provide wireless connectivity in routers with Wi-Fi.

C. Wireless Access Point



Important ports/components:

- **Ethernet:** Connects the AP to the wired network.
- **Power:** Supplies electricity.

Task – 3 : Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

Solution : How Network Devices Help When Using Zomato

Let's use **Zomato** as the example.

When you open Zomato on your phone, your **access point** provides the Wi-Fi connection between your phone and the local network. The **router** then forwards your traffic from your home network toward the Internet and eventually to Zomato's servers. A **firewall** can inspect and filter network traffic according to its security rules before traffic enters or leaves the network.

 Your Phone

|

| Wi-Fi



 Access Point

|

| Ethernet



 Router / Firewall

|

| Internet



 Zomato Servers

Task – 4 : Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help. Hint : Think about device capacity, coverage, and traffic handling.

Solution : **College Wi-Fi Is Slow and Disconnects**

Possible reason

One possible reason is that the **access point is overloaded** because too many students are connected to it simultaneously. An access point has limits on the number of clients and the amount of traffic it can handle.

For example, if hundreds of students are connected to a small number of older access points, the available wireless bandwidth has to be shared between many devices. This can result in slow speeds and intermittent disconnections.

Possible solution

The college could upgrade to **higher-capacity modern access points** and deploy additional APs to improve coverage and distribute users across multiple access points.